

UESL Measurement Protocol: Checklist

User-Experienced Sound Level for wearable robots · Version 1.0 · Bons, Thiede & Rouse, “Sound Matters: Standardizing Audible Noise Reporting for Wearable Robots” (in preparation)

Objective

Quantify device sound during standard operation: practicable, reproducible, ecologically valid.

Target devices

Wearable assistive technology, including (but not limited to) prostheses & exoskeletons.

Metric

A-weighted equivalent continuous sound level, $L_{eq,A}$ (dBA), at the user's ear during use.

1 Equipment & instrumentation

	Sound level meter (SLM)	Binaural sensor unit (BSU)
Example	PCE 322A (PCE Instruments)	SQobold + BHS II (HEAD acoustics)
Cost / access	Low; widely available	Higher
Output	Sound level only	Full audio (~48 kHz): $L_{eq,A}$ + psychoacoustics
In use	Tether to a laptop for ≥ 8 Hz logging	Worn as a normal headset

- ☐ **SLM:** calibrated Class 2 device, **A-weighting**, **FAST** (125 ms), recording at **10 Hz** (≥ 8 Hz avoids gaps in the FAST filter).
 - Microphone extension cable, if used, in place during calibration.
 - Head strap & mount to hold the mic at ear level (e.g., GoPro Head Strap 2.0).
- ☐ **BSU:** calibrated device (e.g., SQobold + BHS II).
- ☐ Calibrate to the standard reference (**94 dB @ 1 kHz**) before each session.

2 Test environment

- ☐ **Absorptive room** (carpet, soft furnishings/curtains) to minimize reverberation.
- ☐ **Ambient** stable and ≥ 6 dB below the expected device noise (ideally < 35 dBA).
- ☐ No significant **correlated sounds** (e.g., treadmill noise while walking).
- ☐ **Floor:** commercial carpet or mat on a solid surface.

3 Subject setup

- ☐ **Mic placement:** SLM at ear level at the **back of the head**; BSU worn over the ears as normal. No hair/clothing rubbing the mic(s).
- ☐ **Device fitting** per manufacturer instructions (CPO where applicable).
- ☐ **Clothing:** nothing loose or noisy (swishing fabrics).
- ☐ **No distance/height correction:** free-field corrections over-correct in real rooms; beyond ~ 0.5 m, ~ 10 cm differences are negligible.

4 Measurement procedure

- ☐ **Task:** cyclic repetition of the intended use, e.g., lower-limb device: walk a large circle at a consistent pace; hand prosthesis: repeatedly pick up & set down a soft, noiseless object.
- ☐ **Each trial:**
 1. User keeps a standard posture and avoids vocalizations.
 2. **Ambient recording:** 10 s, everyone still and silent.
 3. Silently invite the user to begin; allow steady-state speed.
 4. **Active recording:** 20 s while the user performs the task.
- ☐ **9 valid trials:** 3 each at self-selected **slow**, **normal**, and **fast** paces.

5 Data analysis

- ☐ Trim segments with extraneous environmental sounds unrelated to the device.
- ☐ Compute L_{eq} for the ambient and active recording of each trial: average the *energy*, not the decibels (constant sampling rate assumed):

$$L_{eq} = 10 \log_{10} \left(\frac{1}{N} \sum_{i=1}^N 10^{L_{p,i}/10} \right)$$

- ☐ **Device level per trial:** if $(L_{active} - L_{ambient}) < 6$ dBA and $L_{active} > 35$ dBA, correct for ambient:

$$L_{device} = 10 \log_{10} (10^{L_{active}/10} - 10^{L_{ambient}/10}) \quad \text{otherwise } L_{device} = L_{active}$$

- ☐ **UESL** = arithmetic mean of L_{device} over all trials & subjects; also compute the standard deviation.
- ☐ *Recommended (BSU):* add psychoacoustic metrics, especially **loudness N_5** (sone; ISO 532-1) and **sharpness S_{10}** (acum; DIN 45692), e.g., MATLAB `acousticLoudness` / `acousticSharpness` or Python `MoSQUITo`.
- ☐ Describe the loud tail with **percentile levels** (L_{10} , L_1), not peaks.

6 Reporting

- ☐ UESL as a **device- & activity-specific** value (e.g., “Open-Source Leg, level-ground walking: 47.7 dBA”).
- ☐ Include the **standard deviation** and **number of trials**.
- ☐ Document **subject mass & height**, trial procedure, **device settings**, environment, and any **deviations**.
- ☐ State the instrument (SLM or BSU); in our data the BSU read ~2.8 dBA above the SLM on average.
- ☐ Note that ambient subtraction is unreliable when ambient approaches the active level or is unsteady.
- ☐ Comply with institutional and ethical guidelines for human-subject research.

Reference values: level-ground walking (N = 7)

Device	UESL (dBA)	SD	n	Device	UESL (dBA)	SD	n
No device	35.5	2.3	72	Ottobock Empower	43.6	2.3	18
VSO – off	37.4	3.0	72	VSO – on	43.9	1.7	72
MBLUE ankle exo	37.6	2.7	72	DESR VSO	44.0	3.0	72
Proteor Kinnex 2.0	38.2	3.4	54	Dephy Sidekick	44.6	3.4	72
Fillauer AllPro	38.6	2.6	18	OSL v2 – ankle	47.7	4.4	54
Ottobock Genium X4	40.1	2.0	18	Össur Power Knee	49.5	2.6	36
Nylon pants	41.5	2.8	72	OSL v2 – knee + ankle	50.6	2.2	18

Preliminary moderate user-impact crossover (N = 7): ≈ 44.4 dBA UESL $\cdot 12.5$ sone $N_5 \cdot 1.95$ acum S_{10} .

Standards: IEC 61672-1:2013 (SLMs) $\cdot$ ISO 1996-1:2016 (L_{eq}) $\cdot$ ISO 3744:2010 (6 dB ambient criterion) $\cdot$ ISO 532-1:2017 (loudness) $\cdot$ DIN 45692:2009 (sharpness).

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